

TED (15) 5113

Reg :No.....

(Revision-2015)

Signature.....

FIFTH SEMESTER DIPLOMA EXAMINATION IN TOOL AND DIE ENGINEERING.

PLASTIC MOULDING TECHNOLOGY

(Time: 3hrs)

(Max: Marks: 100)

PART-A

(Maximum marks: 10)

I. Answer the following questions in one or two sentences. Each question carries 2 marks.

1. Define thermo set plastic.

2. Classify the injection moulding machine.

3. Define shrinkage.

4. What is multi day light mould?

5. Define bulk factor.

(2x5=10)

PART-B

II. Answer any five of the following questions. Each question carries six marks.

1. State the properties of plastic.

2. Compare the difference between compression moulding and injection moulding.

3. State the function of inserts, and summarise core insert and cavity insert.

4. Illustrate the feed system in an injection mould and write the function of sprue bush,

Gate and runner.

5. Discuss the advantages of hot runner injection mould.

6. Summarise an external undercut and justify the necessity of splits used for moulding

External undercut compound.

7. State the benefits of plastic decorating.

(5x6=30)

PART-C

(Maximum marks:60)

(Answer one full question from each unit. Each full question carries 15 marks)

III .

- a) Draw the block diagram of hot runner injection mould and write its essentiality. (8)
- b) Write the purpose for which the following are used in plastic. (7)
 - i) Accelerator ii)Lubricants iii)Stabilizer iv)Plasticizer

OR

IV .

- a) Identify the various parts of extrusion moulding machine with neat sketch. (8)
- b) Explain the working of plunger type transfer moulding machine. (7)

UNIT-II

V.

- a) Define parting surface. Explain flat parting surface with suitable sketch. (8)
- b) Draw a simple sketch of integer cavity plate cooling and write its necessity. (7)

OR

VI.

- a) Explain the working principle of two stage screw type injection moulding. (8)
- b) State the function of bolsters. Explain any two type of bolster with sketch. (7)

UNIT-III

VII.

- a) Discuss the advantages and disadvantages of insulated runner (runner less) mould system. (8)
- b) Identify the various parts of Triple daylight mould with sketch. (7)

OR

VIII.

- a) Describe the method of operation of finger can actuation of splits with sketch. (8)
- b) Explain the basic principle of side core with suitable figure. (7)

UNIT-IV

IX.

- a) Sketch and explain the working of Axial (spider) flow head used in blow moulding (8)
- b) Write short notes on i) stamp decorating ii) In mould decorating. (7)

OR

X.

- a) Briefly describe the method of calculating the depth of loading chamber in a compression mould. (8)
- b) Illustrate the method of vacuum metalizing. (7)

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PLASTIC MOULDING TECHNOLOGY

(Maximum Marks: 100)

Time:3hrs

PART-A

I. (Answer the following questions in one or two sentences. Each question carries two marks.)

1. Define thermo plastics.
2. List the types of compression moulding.
3. State the function of bolster plates.
4. Define the term single daylight mould.
5. What is flash allowance?.

(2x5=10)

PART-B

II. (Answer any five of the following questions. Each question carries six marks.)

- 1) Indicate the function of Fillers and Stabilizers in plastics as additives.
- 2) Briefly describe the method of thermo forming with sketch.
- 3) Discuss the fundamental requirements of bolsters.
- 4) State the relations between Core, Cavity and Impression.
- 5) Identify the limitations of hot runner.
- 6) Draw a multy daylight mould and compare it with a single day light mould.
- 7) Sketch any two types of Ribs and discuss how it effects the component?

(6x5=30)

PART-C

UNIT-I

(Maximum marks: 60)

III.

a) Describe the following compression mould with neat sketch.

i) Positive mould; ii) Landed positive mould. (8)

b) Generate a method of operation of injection moulding to make a two coloured plastic cup.(7)

OR

IV.

a) Illustrate the working principle of Reaction injection mould. (8)

b) Describe the working principle of Rotational moulding process. (7)

UNIT-II

V.

a) Explain component making by the method of two stage plunger type injection moulding . (8)

b) Define the following.

i)Form pin, ii) Sprue bush, iii) Register ring. (7)

OR

VI.

a) List the different types of gate and write the comments on sprue (center) gate. (8)

b) State the importance of sleeve ejection and draw its figure . (7)

UNIT-III

VII.

a) Mark the various parts of a fixed threaded core and write its advantages. (8)

b) Explain the basic principle of side cavity with suitable figure. (7)

OR

VIII.

- a) Sketch and explain the principle of operation of Dog leg cam. (8)
- b) Illustrate the method of operation of splits by compression spring method. (7)

UNIT-IV

IX.

- a) Explain the method coating by flame spraying. (8)
- b) Briefly describe the factors to find out the parison wall thickness. (7)

OR

X.

- a) How a plastic component coating by a dipping method, explain with figure . (8)
- b) Write the comments on runner dimensions and gate dimensions. (7)
